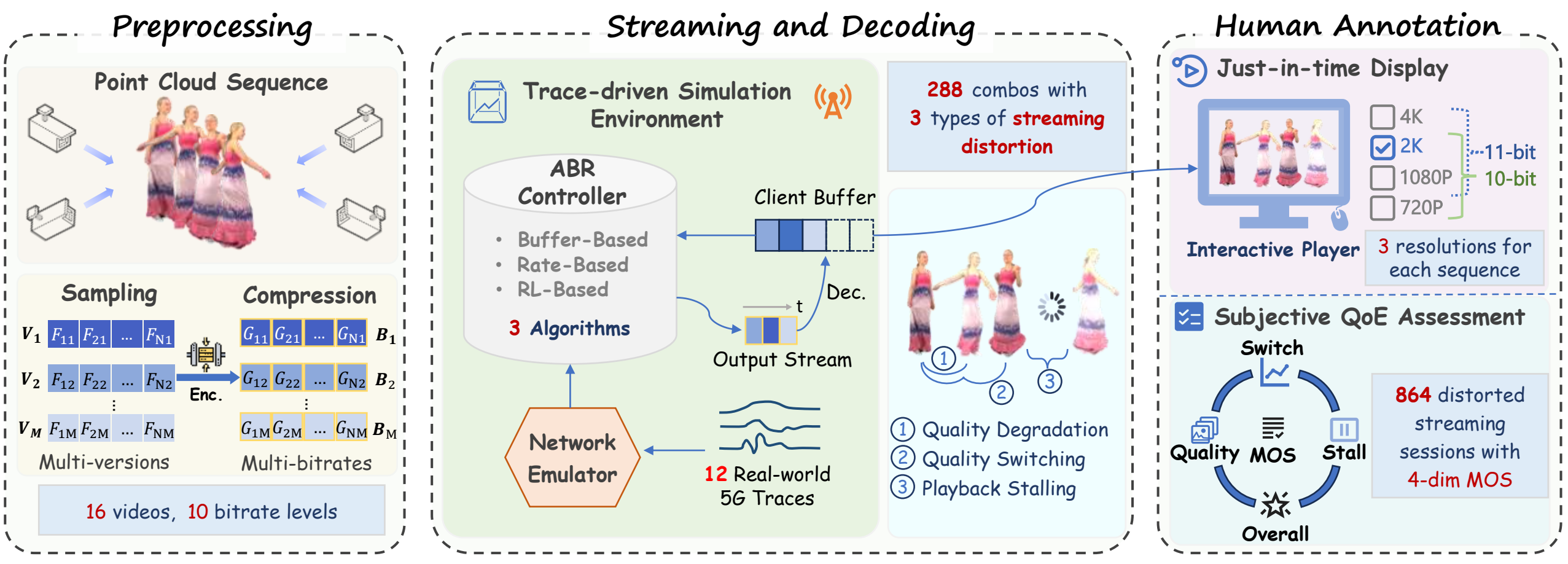
Preprocessing
Point Cloud Sequence
Sampling
Compression
V_1 F_{11} F_{21} ... F_{N1}
V_2 F_{12} F_{22} ... F_{N2}
V_M F_{1M} F_{2M} ... F_{NM}
Enc.
G_{11} G_{21} ... G_{N1} B_1
G_{12} G_{22} ... G_{N2} B_2
G_{1M} G_{2M} ... G_{NM} B_M
Multi-versions
Multi-bitrates
16 videos, 10 bitrate levels
Streaming and Decoding
Trace-driven Simulation Environment
ABR Controller
Buffer-Based
Rate-Based
RL-Based
3 Algorithms
Client Buffer
Dec.
t
Output Stream
Network Emulator
12 Real-world 5G Traces
288 combos with 3 types of streaming distortion
① Quality Degradation
② Quality Switching
③ Playback Stalling
Human Annotation
Just-in-time Display
4K
2K
1080P
720P
11-bit
10-bit
Interactive Player
3 resolutions for each sequence
Subjective QoE Assessment
Switch
Quality
MOS
Stall
Overall
864 distorted streaming sessions with 4-dim MOS